

Ben Chung

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EDUCATION

University of British Columbia

Bachelor of Applied Science – Electrical and Computer Engineering

Expected Grad: Nov 2027

EXPERIENCE

Junior Electrical Designer

September 2025 – May 2026

Smith + Andersen

- Designed SLDs, power, and lighting layouts for tenant buildings using AutoCAD MEP, Revit, and Visual Lighting.
- Wrote EPRs to quantify electrifying existing equipment and implement EV charging. Determined electrical demand, spare capacity, and optimized power efficiency.
- Performed calculations to size feeders, circuit breakers, mech equipment, and panels according to BCBC and CEC.

Airside Project Management

January 2025 – August 2025

Vancouver Airport Authority (YVR)

- Supported YVR's \$170M North Runway Program by assisting with technical coordination, project documentation, and stakeholder communication across paving, drainage, and electrical work scopes.
- Created a master project tracker based on P6 schedule to determine earned value of completed work and modelled progress rates to extrapolate forecasted project completion dates.
- Created a manual to familiarize new users with YVR's Airfield Lighting System and Power Distribution System with adherence to TP312 aerodrome compliance standards.

PROJECTS

Self-Balancing Robot (2016 Hoverboard) | *C++, Python, Control Systems, Electronics, Embedded Programming*

- Developed embedded C++ control software using a cascaded PID architecture, where the outer position PID adjusts target angle while the inner angle PID loop controls motor PWM output.
- Fine-tuned gyroscope and accelerometer filter values, PID gains, and encoder & motor response characteristics for precise robot balance and control.
- Integrated BLE connection with a Flutter mobile app to control robot movements.

iPod Machine with Volume Indicator | *SystemVerilog, SignalTap, Embedded Programming*

- Designed a keyboard-input FSM to control flash memory writes. Designed a separate FSM to synchronize address controller and flash memory reads.
- Programmed an interrupt routine to sample and real-time filter audio data from flash memory to visualize audio strength using an embedded PicoBlaze microcontroller.
- Verified system functionality using waveform simulations and SignalTap.

Omegle Chat Room | *Python, RTOS, TCP/IP*

- Built a multi-client and server-hosting TCP chat application using Python sockets, enabling clients and servers to exchange messages in real time.
- Implemented disconnect detection, status broadcasts, socket cleanup, and mutex lock to preserve message order and prevent race conditions.

RC4 Brute-Force Message Decryptor | *SystemVerilog, SignalTap, FPGA*

- Designed a 16-state FSM decryptor with pipelined reads and a start-finish controller communication protocol.
- Integrated RAM and ROM modules to read and write key scheduling, ciphertext storage, and decrypted output writes.
- Parallelized the brute-force key search across four processing cores to accelerate search throughput.

TECHNICAL SKILLS

Programming/HDL Languages: Python, C/C++, Java, SystemVerilog, Assembly

Proficient Programs: LTSpice, Matlab, Visual Studio, Git, AutoCAD, Revit, Quartus, SignalTap, MS365, Bluebeam

Awards: UBC Centennial Entrance Award, BC Achievement Scholarship

Hobbies: Boxing, Hiking, Volleyball